

CNT Experiments

Marco Solime, NII, 4 March 2026





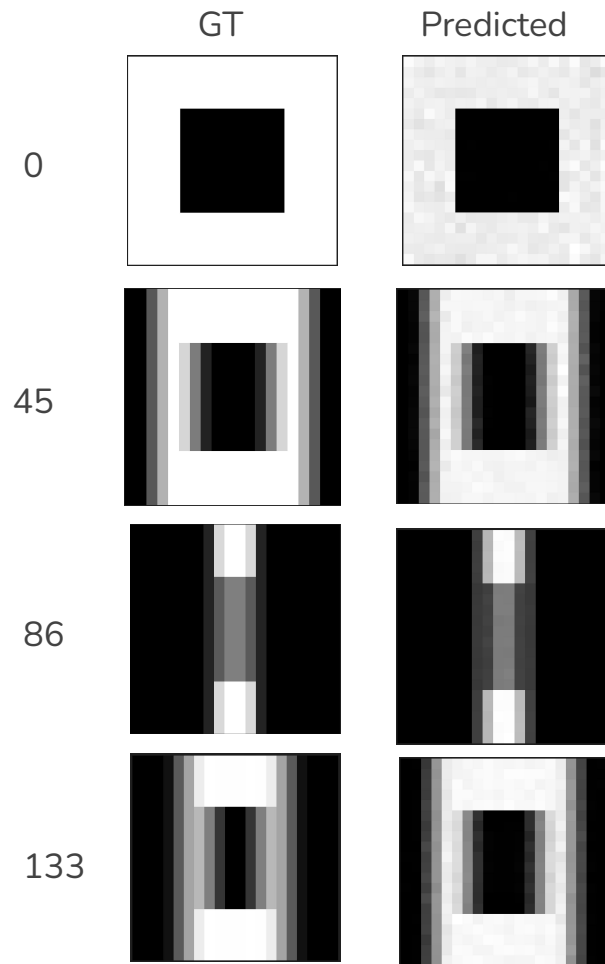
Experiment 1 — Thin Slice

- Non-empty volume, size 4x20x20, with squared hole
- Detector size, 20x20x20
- Number of projections: 360
- **Good** 2D predictions and 3D reconstruction
- SaxNerf can reconstruct **thin slices** if it is given **void boundaries**



GT

Predicted





Experiment 2 – Reducing n. of projections

Projs

GT

Predicted

180



90



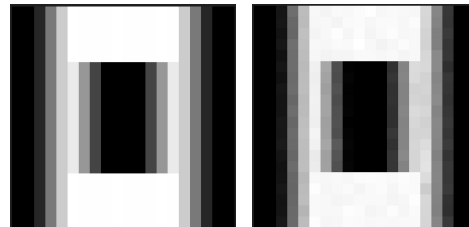
45



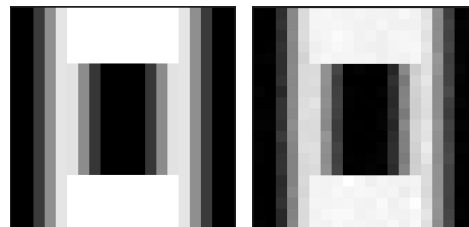
GT

Predicted

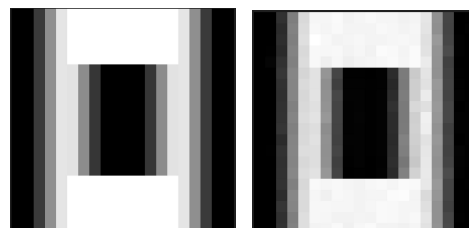
Proj: 25



Proj: 12



Proj: 6





Experiment 2 – Reducing n. of projections

Projs

GT

Predicted

22



11



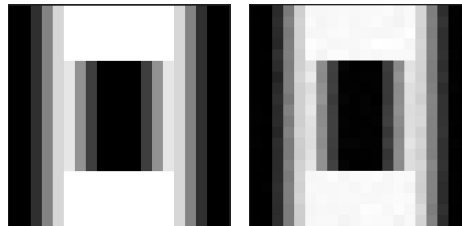
5



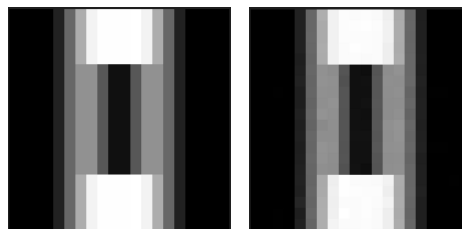
GT

Predicted

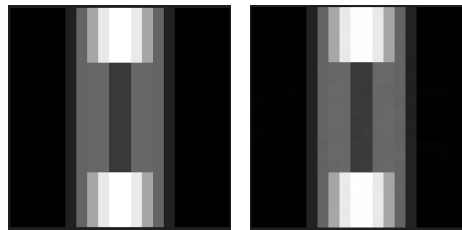
Proj: 3



Proj: 2



Proj: 1





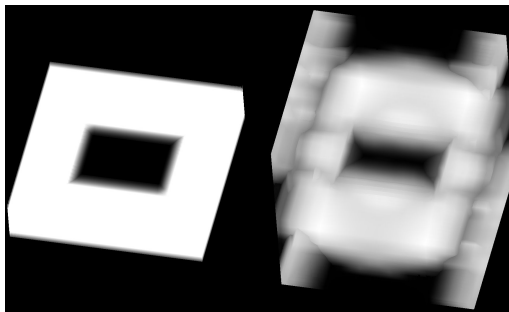
Experiment 2 – Reducing n. of projections

Projs

GT

Predicted

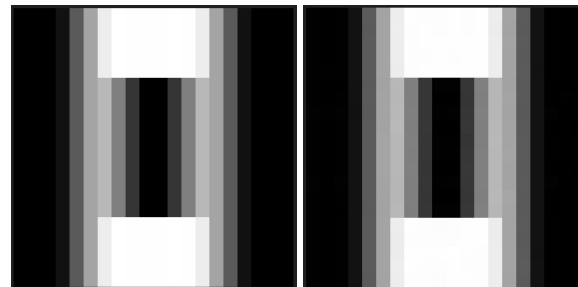
3



GT

Predicted

Proj: 1





Data from CT imaging — 1,310 nm

Angle

0



Angle

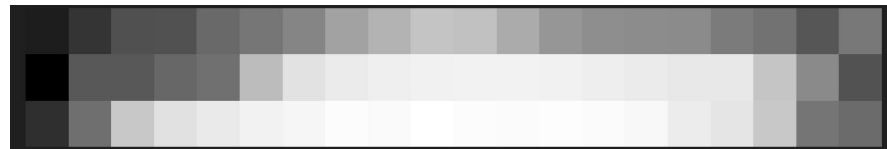
179



89



269



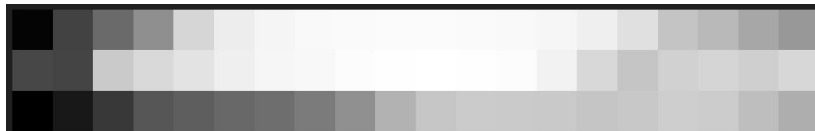


Data from CT imaging — 10.3 μm

0



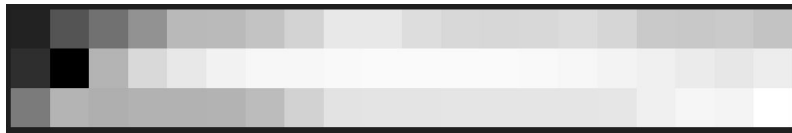
179



89



269



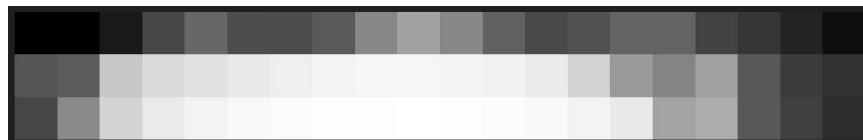


Data from CT imaging — 976 nm

0



179



89



269

